STIM Rev 1 Channel

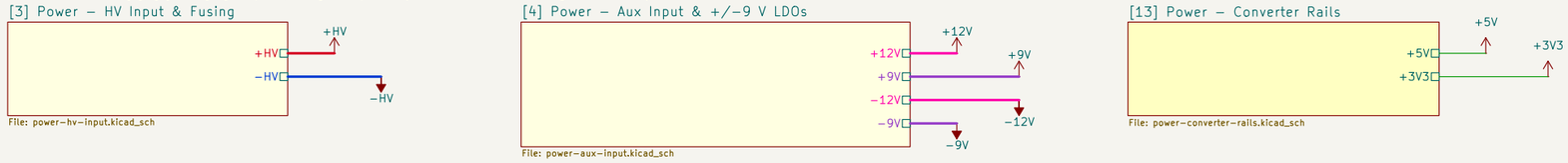
28-Oct-2026

Variant: Preliminary

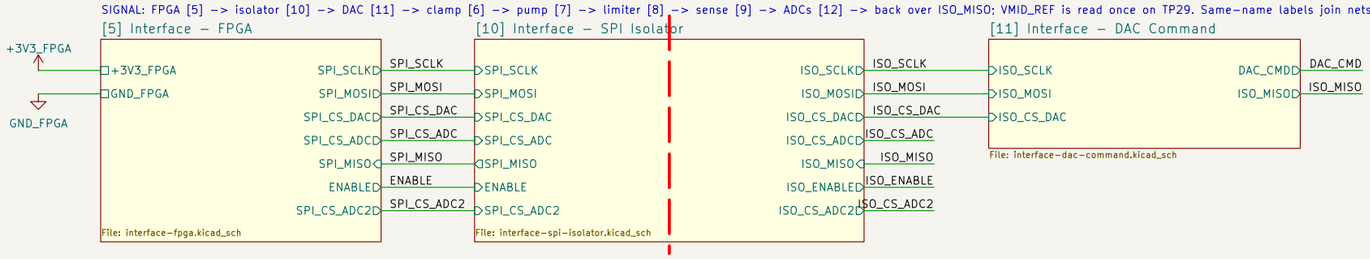
Rev 2.1.0-dev

+/-100 V INA149 Howland current-pump stimulator channel: minimal analog path for bench bring-up (UBC ELEC 491, JY063)

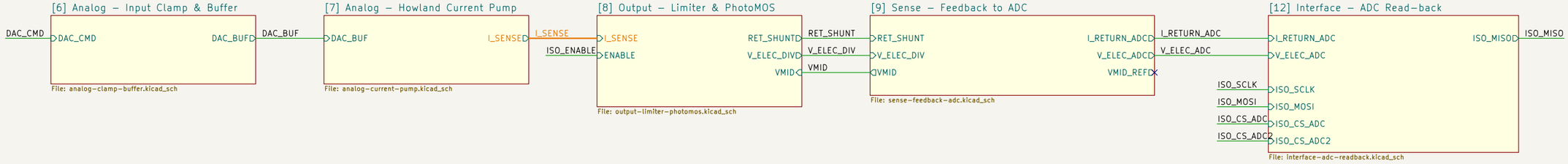
POWER: rails leave [3], [4] and [13] as ports; the global power symbols live here. GND is global on every sheet.



SIGNAL: FPGA [5] -> isolator [10] -> DAC [11] -> clamp [6] -> pump [7] -> limiter [8] -> sense [9] -> ADCs [12] -> back over ISO_MISO; VMID_REF is read once on TP29. Same-name labels join nets.



ISOLATION BARRIER (inside [10]): left of it J3, +3V3_FPGA, GND_FPGA (FPGA side); right of it the board, GND. Spacing rule set with the isolator part.



DESIGN NOTE:
How and why a circuit is sized (grey).

LAYOUT NOTE:
Placement and routing rules from the PCB design reference (blue).

MISC NOTE:
Bench, test and procurement notes (orange). A tag such as D1, L2 or M1 next to a circuit points to the note with that number.



H1 M3 3.4mm NPTH

H2 M3 3.4mm NPTH

H3 M3 3.4mm NPTH

H4 M3 3.4mm NPTH

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Doc - Cover	File Name: STIM_Rev1.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 13

[2] Doc - Revision History

23-OCT-2026 - Initial Capture
Version: 1.0.0 Draft

- Captured the full Rev 1 channel from the simulated netlist and the P4 BoM: 14 sheets, 156 parts.

- Included the full safety chain: over-current and mismatch trips, fault latch, power-on reset, rail monitors, parking switch.

- Netlist matched expected_netlist.csv with 0 differences; ERC clean.

25-OCT-2026 - Minimal Rev 1
Version: 2.0.0-dev Preliminary

- Cut the board to the analog path for bench bring-up: HV input, clamp, buffer, Howland pump, limiter, PhotoMOS, TVS, read-back.

- Removed the trips, latch, power-on reset, rail monitors, enable logic and parking switch (kept in simulation for Rev 2).

- Changed rail shunts RSP/RSN to 10 Ohm for current measurement.

- Changed FPGA connector J3 to DAC_CMD, ENABLE, I_RETURN_ADC, V_ELEC_ADC, VMID_REF and grounds; no power.

- Added the read-back stage (OPA4192) for the FPGA ADC.

- Added bench bias jumpers JP1/JP2, current-loop header JP3, 29 test loops and M3 mounting holes.

26-OCT-2026 - Schematic Review
Version: 2.0.0-dev Preliminary

- Reduced the low-voltage input to one +12 V (J2, 2-pin screw terminal); U32 isolated DC-DC makes +/-12 V on board.

- Removed +5 V; LM4040 bias R50 now from +9 V (5.1k).

- Changed U1 and U3 from OPA171 to OPA192: output accuracy passes without the zero calibration.

- Added red LEDs: PWR (D8), HV LIVE (D7) and OUT EN (D6, in the PhotoMOS LED string; RLED1 now 750 Ohm).

- Fixed the AQY214S symbol (its pins were hidden).

- Resized passives: 0603 by default; 0805/1206/2512 only where voltage, power or precision needs it; one part per value.

- Wired the top sheet; ERC clean; page numbers now follow the sheet titles.

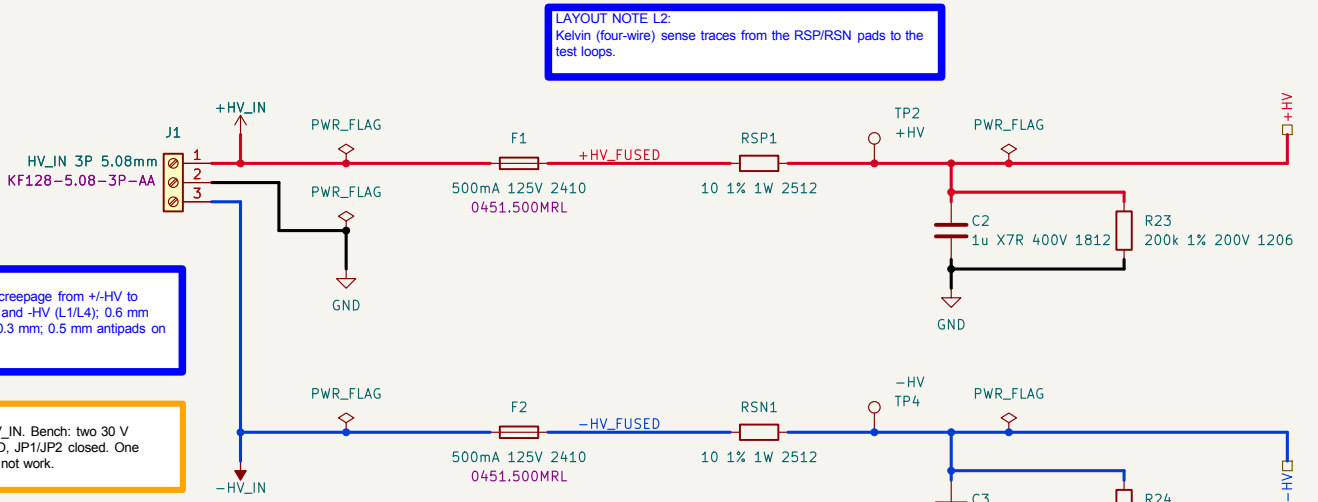
- Formatted after the FoC Motor Controller and AMULET: cover index, block frames with captions, this page. PCB text in JetBrains Mono; 3D model for every part.

DD-MMM-YYYY - Next Revision
Version: _____

- _____
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Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Doc - Revision History	File Name: revision-history.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[2] Doc - Revision History/		Reviewer:	Size: A4	Sheet: 2 of 13

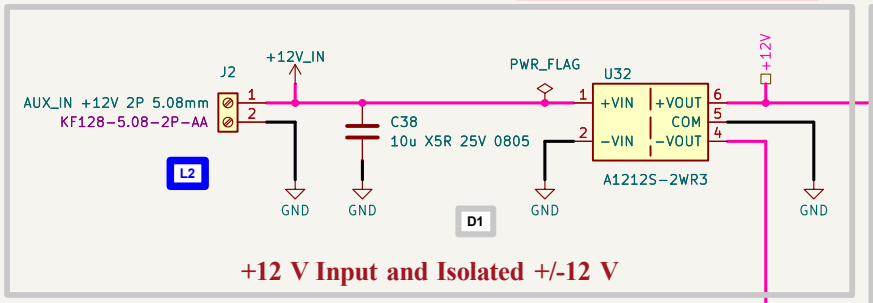
[3] Power - HV Input & Fusing



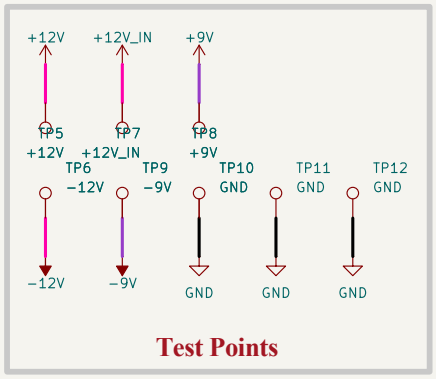
Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Power - HV Input & Fusing	File Name: power-hv-input.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[3] Power - HV Input & Fusing/		Reviewer:	Size: **A4**	Sheet: **3** of **13**

[4] Power - Aux Input & +/-9 V LDOs

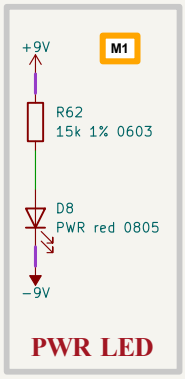
CONFIRM GND AND ISO_GND



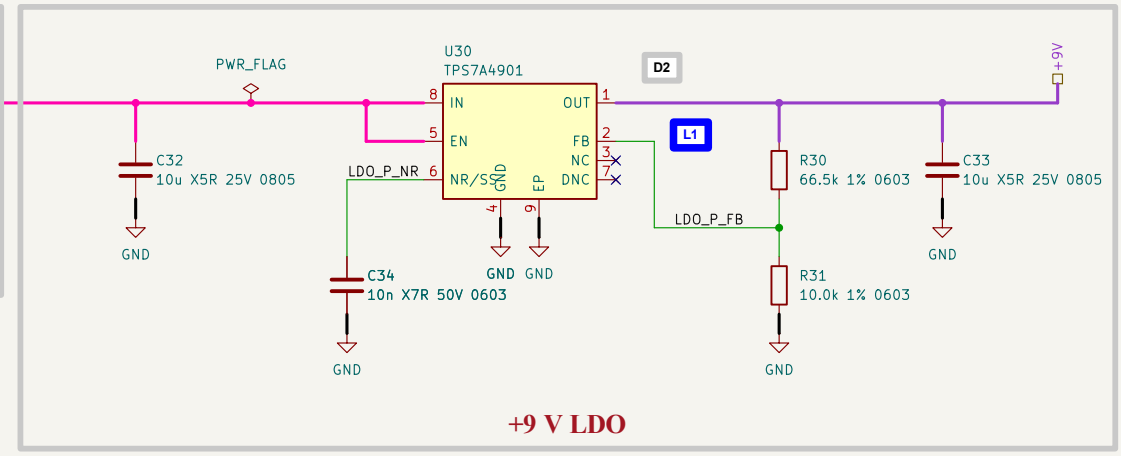
+12 V Input and Isolated +/-12 V



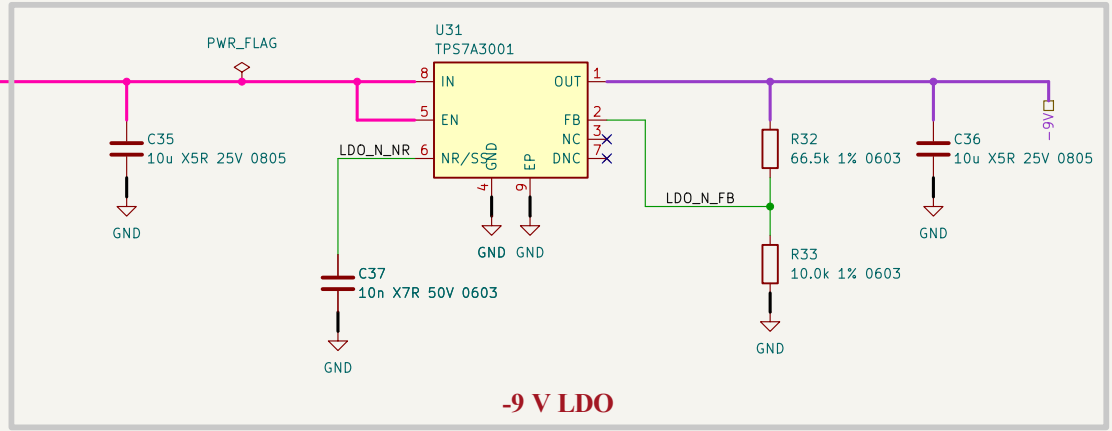
Test Points



PWR LED



+9 V LDO



-9 V LDO

DESIGN NOTE D1:
U32 A1212S-2WR3 (YLPTEC): input 10.8-13.2 V. Pinout per the YLPTEC datasheet: check the module label. The -12 V load (~7 mA) is under the 10 % minimum load, so that rail may sit up to 10 % high: fine for U31 and U22.

DESIGN NOTE D2:
+/-9 V sets the buffer ceiling (D-01): 1.188 V x (1 + 66.5k / 10k) = 9.09 V (9.00-9.27 V). Changing R30-R33 changes the zener-open current ceiling (9.15 mA): re-check D-01 first.

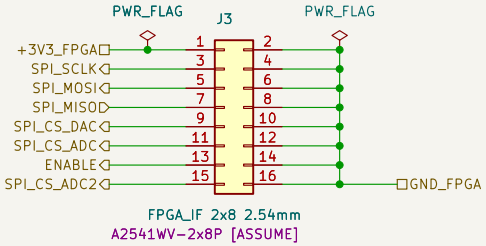
LAYOUT NOTE L1:
Thermal pad (pin 9) to GND with vias; CIN, COUT and the NR/SS cap beside their pins. 0.65 mm pitch: 0.15 mm pad gaps (rule "fine-pitch LDO pads"), mask bridges allowed.

LAYOUT NOTE L2:
J2 and U32 on the top edge (LV side); C38 at the U32 input pins.

MISC NOTE M1:
PWR LED (D8) lights only when +12V_IN, U32 and both LDOs are up: the first check at power-up.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Power - Aux Input & +/-9 V LDOs	File Name: power-aux-input.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[4] Power - Aux Input & +/-9 V LDOs/		Reviewer:	Size: A4	Sheet: 4 of 13

[5] Interface - FPGA



DESIGN NOTE D1:
J3 2x8 IDC: 1 +3V3_FPGA, 3 SPI_SCLK, 5 SPI_MOSI, 7 SPI_MISO, 9 SPI_CS_DAC, 11 SPI_CS_ADC, 13 ENABLE (3.3 V; low or open = switches open), 15 SPI_CS_ADC2 (3.3 V, active low: chip select of the second ADC U42 on [12] through U44 on [10]; no spare pin left, see [10] D3); even pins GND_FPGA. All nets here are FPGA-side; they cross to the board only through [10]. [ASSUME] +3V3_FPGA from the FPGA powers isolator side 1 (ICD_J3 said no power on J3: decide and update the ICD). J5 on [10] is the board-side twin of this header for the isolator bypass. Analog DAC_CMD / read-back pins are gone: [11] and [12] do that.

LAYOUT NOTE L1:
J3 on the left (LV) edge, on the FPGA side of the isolation barrier with side 1 of [10]. Every signal has a GND_FPGA neighbour on the ribbon. GND_FPGA is its own pour under J3 and the isolator side 1, not board GND.

MISC NOTE M1:
FPGA: hold the DAC at 0 for 3 ms after ENABLE (D-08; the AQY214S closes in 0.5 ms, so this can shorten); clear the DAC when ENABLE drops; one SPI bus, three chip selects (DAC [11], two ADCs [12]); subtract the once-measured VMID_REF (TP29) from both read-back channels in software.

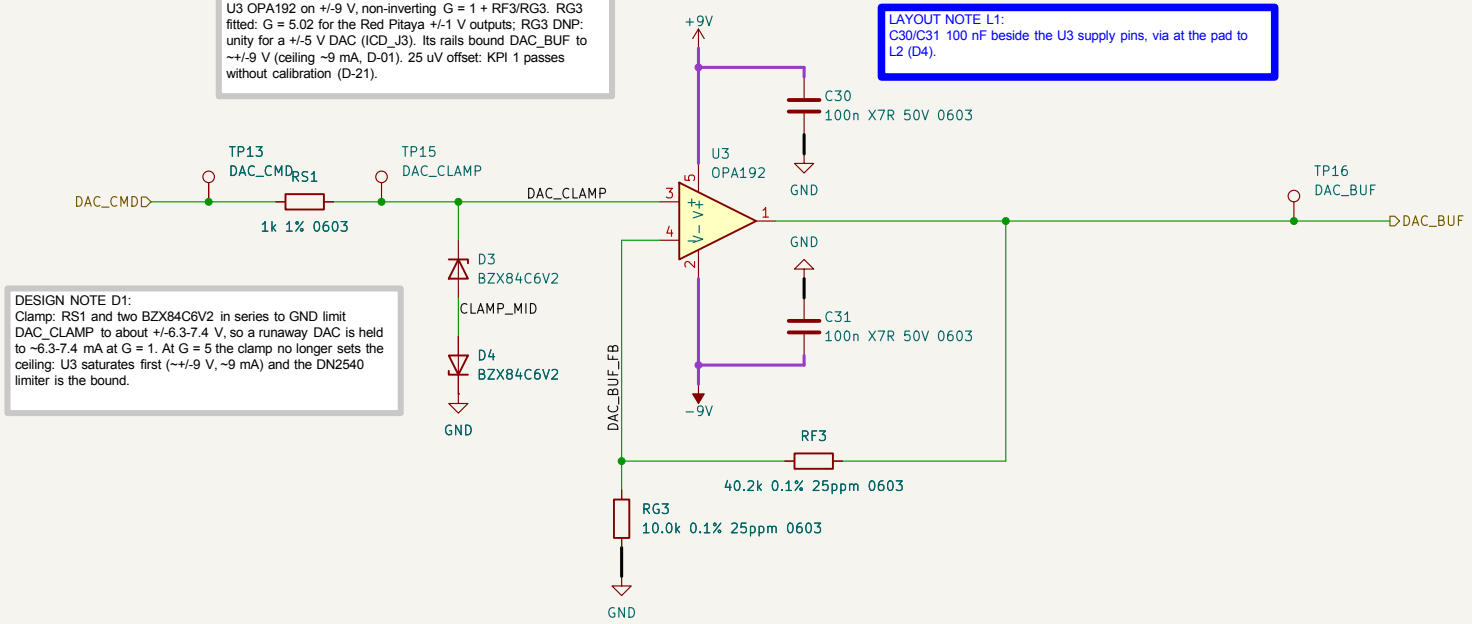
Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - FPGA	File Name: interface-fpga.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[5] Interface - FPGA/		Reviewer:	Size: **A4**	Sheet: **5** of **13**

[6] Analog - Input Clamp & Buffer

MISC NOTE M1:
TP13 DAC_CMD -> TP15 DAC_CLAMP -> TP16 DAC_BUF: probe all three to see the clamp act.

DESIGN NOTE D2:
U3 OPA192 on +/-9 V, non-inverting G = 1 + RF3/RG3. RG3 fitted: G = 5.02 for the Red Pitaya +/-1 V outputs; RG3 DNP: unity for a +/-5 V DAC (ICD_J3). Its rails bound DAC_BUF to ~+/-9 V (ceiling ~9 mA, D-01). 25 uV offset: KPI 1 passes without calibration (D-21).

LAYOUT NOTE L1:
C30/C31 100 nF beside the U3 supply pins, via at the pad to L2 (D4).



DESIGN NOTE D1:
Clamp: RS1 and two BZX84C6V2 in series to GND limit DAC_CLAMP to about +/-6.3-7.4 V, so a runaway DAC is held to ~6.3-7.4 mA at G = 1. At G = 5 the clamp no longer sets the ceiling: U3 saturates first (~+/-9 V, ~9 mA) and the DN2540 limiter is the bound.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Analog - Input Clamp & Buffer	File Name: analog-clamp-buffer.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[6] Analog - Input Clamp & Buffer/		Reviewer:	Size: A4	Sheet: 6 of 13

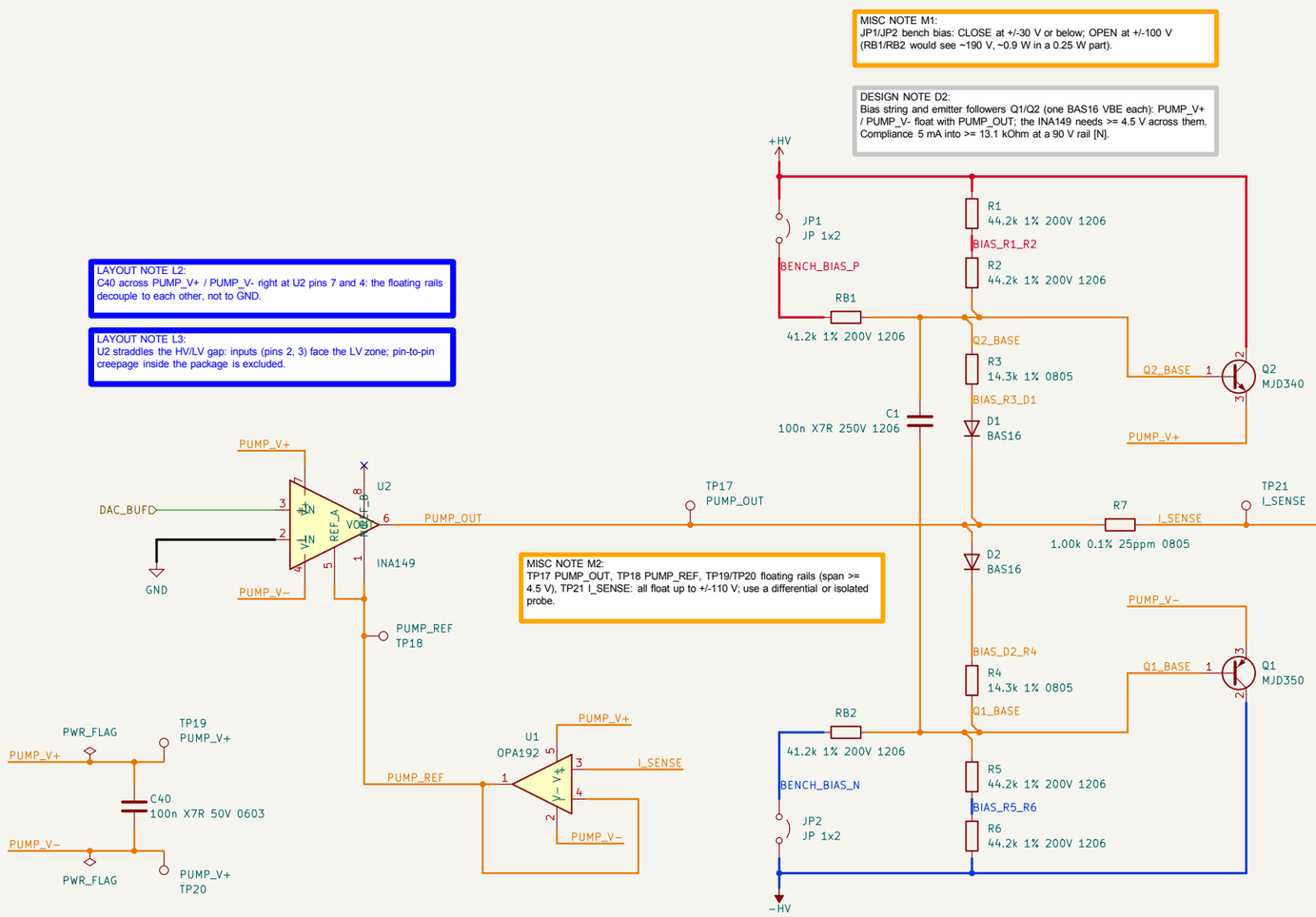
[7] Analog - Howland Current Pump

LAYOUT NOTE L2:
C40 across PUMP_V+ / PUMP_V- right at U2 pins 7 and 4: the floating rails decouple to each other, not to GND.

LAYOUT NOTE L3:
U2 straddles the HV/LV gap: inputs (pins 2, 3) face the LV zone; pin-to-pin creepage inside the package is excluded.

MISC NOTE M1:
JP1/JP2 bench bias: CLOSE at +/-30 V or below; OPEN at +/-100 V (RB1/RB2 would see ~190 V, ~0.9 W in a 0.25 W part).

DESIGN NOTE D2:
Bias string and emitter followers Q1/Q2 (one BAS16 VBE each): PUMP_V+ / PUMP_V- float with PUMP_OUT; the INA149 needs >= 4.5 V across them. Compliance 5 mA into >= 13.1 kOhm at a 90 V rail [N].

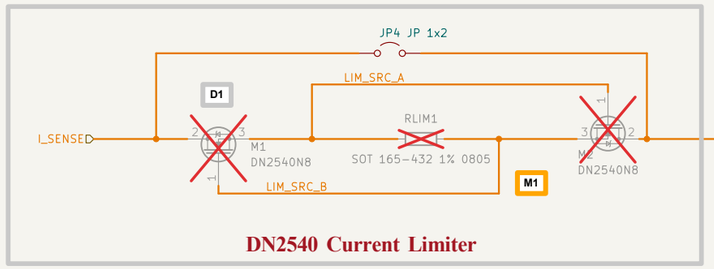


DESIGN NOTE D1:
1 mA/V set by R7 = 1.00 k 0.1 % 25 ppm (0805 for self-heating). U2 INA149 senses across R7; U1 OPA192 buffers I_SENSE back to the REF pins (D-05).

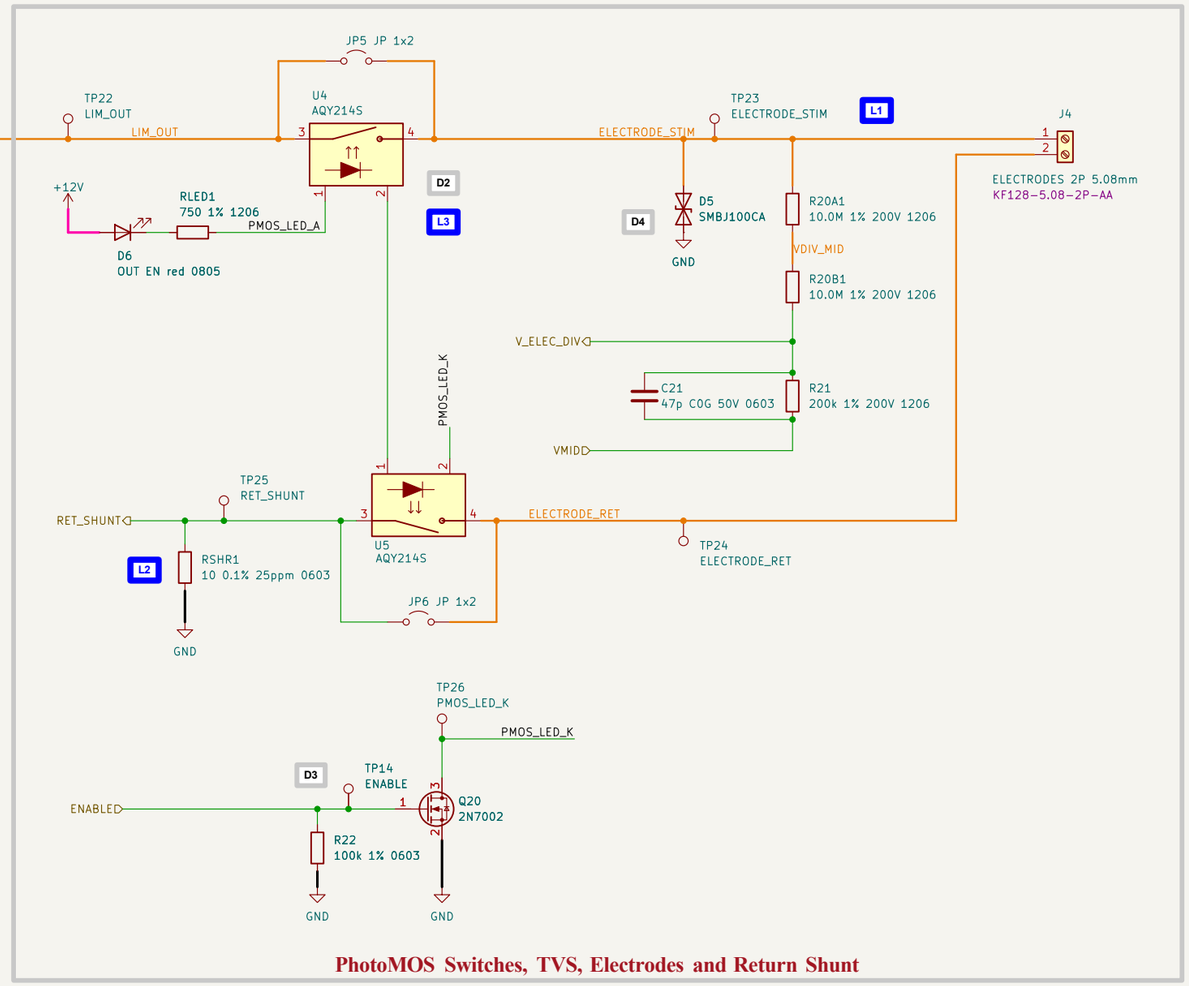
LAYOUT NOTE L1:
Q1/Q2 DPAK: >= 300 mm^2 L1 pour on each collector tab (D6; up to 1.1 W in a 10 mA fault), 1.5 mm to everything; no thermal vias into L2.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Analog - Howland Current Pump	File Name: analog-current-pump.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[7] Analog - Howland Current Pump/		Reviewer:	Size: **A3**	Sheet: **7** of **13**

[8] Output - Limiter & PhotoMOS



DN2540 Current Limiter



PhotoMOS Switches, TVS, Electrodes and Return Shunt

DESIGN NOTE D1:
Limiter (D-02): matched DN2540 pair, one shared RLIM selected on test for 7.6 mA: R = (|Vgs at 7.6 mA| + 10 mV) / 7.6 mA, nearest E96. Limit 6.4-8.8 mA over spread and 0-50 C.

DESIGN NOTE D2:
U4 (stimulating) and U5 (return) share one LED string: one failure opens both. D6 OUT EN is in the string and lights only while the electrodes are live; RLED1 750 Ohm sets ~10 mA.

DESIGN NOTE D3:
ENABLE low or open (R22 pull-down) = LED string off = both switches open.

DESIGN NOTE D4:
D5 SMBJ100CA: VRWM 100 V, VBR 111 V min, above the 110 V rail.

LAYOUT NOTE L1:
Divider: R20A1 in the HV island, R20B1 straddles the gap (V_ELEC_DIV pad on the LV side), R21/C21 on the LV side next to it.

LAYOUT NOTE L2:
RSHR: Kelvin traces from its pads to U22 (0.1 % part, D5); keep the return path J4 -> U5 -> RSHR -> GND short.

LAYOUT NOTE L3:
PhotoMOS straddle the gap: LED pins (1, 2) toward the LV zone, switch pins (3, 4) toward HV.

MISC NOTE M1:
Buy ~10 DN2540N8 per channel; pair them within 100 mV of Vgs at 7.6 mA. RLIM1 is 0805 for hand rework.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Output - Limiter & PhotoMOS	File Name: output-limiter-photomos.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[8] Output - Limiter & PhotoMOS/		Reviewer:	Size: **A3**	Sheet: **8** of **13**

[9] Sense - Feedback to ADC

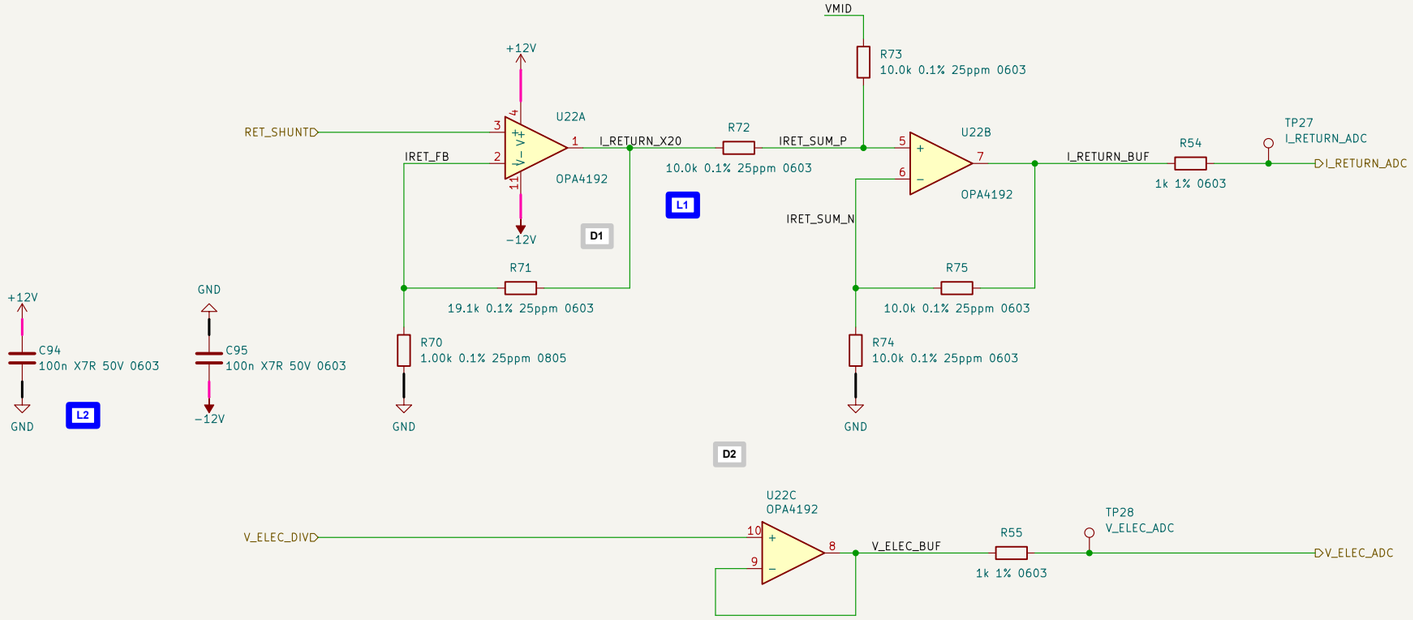
DESIGN NOTE D1:
I_RETURN_ADC = VMID + 0.201 V/mA: RSHR 10 mV/mA x 20.1 (U22A, R71/R70) + VMID (U22B, R72-R75 0.1 %). +/-5 mA -> 0.66-2.66 V.

DESIGN NOTE D2:
V_ELEC_ADC = VMID + V_elec / 101 (20 MOhm / 200 k returned to VMID); U22C buffers it (20 pA bias).

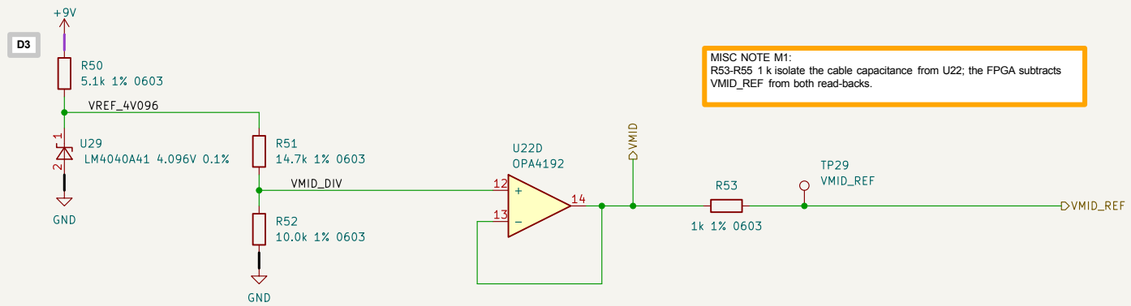
DESIGN NOTE D3:
VMID = 4.096 V x 10.0k / 24.7k = 1.658 V (half of a 3.3 V ADC). Other ADC range: re-select R51/R52. U29 runs at ~0.8 mA from +9 V.

LAYOUT NOTE L1:
0.1 % set R72-R75: short, symmetric traces; no copper pour between their pads.

LAYOUT NOTE L2:
C94/C95 at U22 pins 4 and 11, via at the pad.



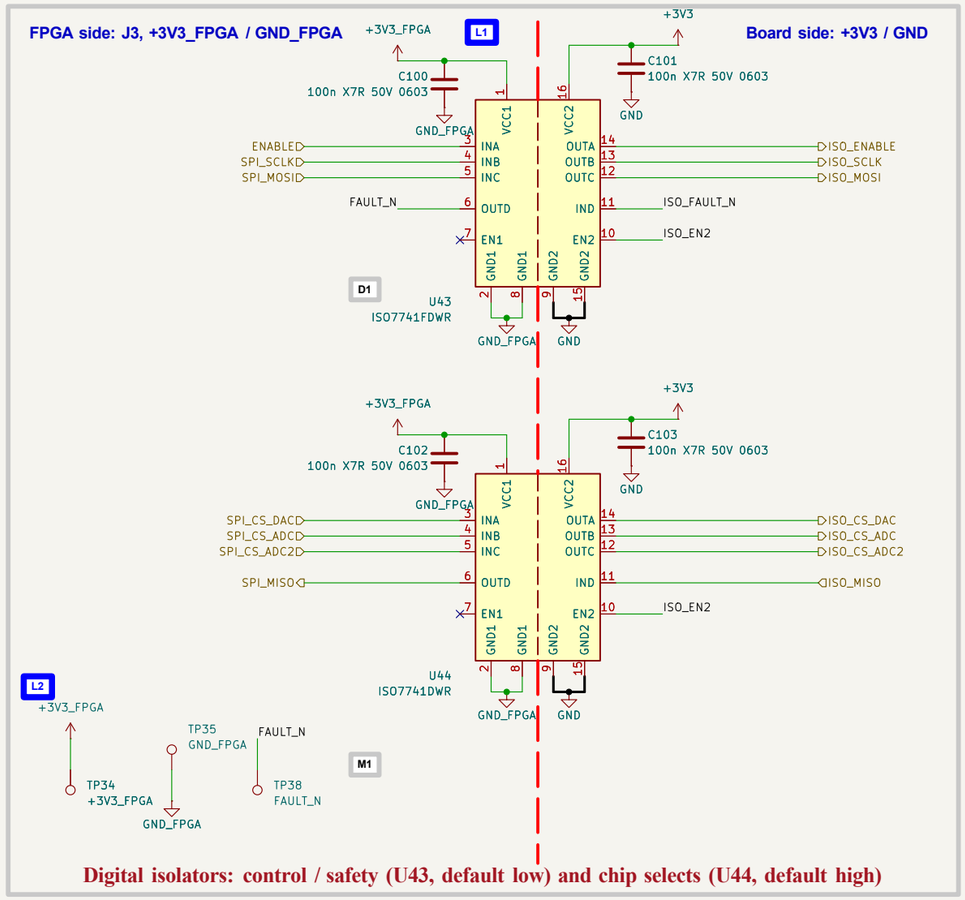
Read-back Amplifiers: Return Current x20 + VMID, Electrode Voltage Buffer

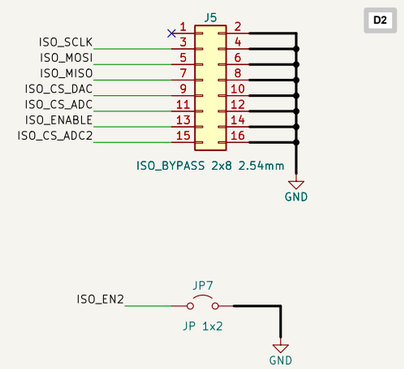


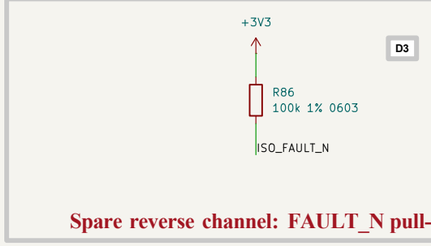
VMID Reference

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Sense - Feedback to ADC	File Name: sense-feedback-adc.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[9] Sense - Feedback to ADC/		Reviewer:	Size: **A3**	Sheet: **9** of **13**

[10] Interface - SPI Isolator







DESIGN NOTE D1:
Two TI ISO7741 (3 forward + 1 reverse channel each, 5 kVrms reinforced, SOIC-16 DW, output enables EN1/EN2). U43 = ISO7741F (default LOW): ENABLE, SCLK, MOSI. U44 = ISO7741 (default HIGH): CS_DAC, CS_ADC, CS_ADC2, MISO back. Six forward lines fit exactly; the CS lines sit on the default-HIGH part so nothing is selected and nothing drives MISO when the FPGA side is unpowered. Side 1 runs from +3V3_FPGA (J3 pin 1), side 2 from +3V3 [13]; both 2.25-5.5 V. One 100 nF at each VCC pin (C100-C103).

DESIGN NOTE D2:
Default states with J3 unplugged or the FPGA unpowered: ISO_SCLK / ISO_MOSI / ISO_ENABLE LOW (PhotoMOS open), ISO_CS_DAC / ISO_CS_ADC / ISO_CS_ADC2 HIGH. EN1 / EN2 open = outputs enabled (internal pull-up, DS pin table). JP7 pulls both EN2 low for the J5 bypass; the side-1 outputs (MISO, FAULT_N) stay enabled, which is fine because J3 is then unplugged.

DESIGN NOTE D3:
Spare: U43 channel D, board -> FPGA, named FAULT_N. Board-side input ISO_FAULT_N is pulled up by R86 (idle = not faulted); the FPGA-side output goes to TP38 only, because J3 has no free pin (15 = CS_ADC2 since the two-ADC decision of 2026-11-08). Rev 2: J3 2x10 for FAULT_N and any PGOOD / OK lines.

LAYOUT NOTE L1:
Isolation barrier under the two isolators: no copper on any layer across the gap except the isolator bodies; GND_FPGA pour under J3 and side 1 only. Spacing per IEC 60664-1 for the working voltage set by the +/-100 V electrode side (hv-insulation rule in STIM_Rev1.kicad_dru); the DW body gives 8 mm creepage. C100-C103 at the VCC pins, via at the pad.

LAYOUT NOTE L2:
J5 next to the isolators on the board side; same orientation as J3 (pin 1 marked) so the ribbon moves straight across. J3 and J5 must not be plugged at the same time with JP7 open.

MISC NOTE M1:
Bring-up: with JP7 OPEN and J3 unplugged, check ISO_ENABLE LOW (TP on [8]) and ISO_CS_* HIGH on J5. Then plug J3: FPGA toggles ENABLE, check ISO_ENABLE follows (DS 11 ns). Fallback if the SPI link misbehaves: JP7 CLOSED, ribbon on J5, FPGA drives the converters directly (3.3 V logic on both sides). Sourcing: ISO7741FDWR is LCSC / JLCPCB only today (DigiKey CA backorder); ISO7742FDWR (C2862403, DigiKey 3352 pcs) is the drop-in if it runs out: same footprint, 2 forward channels, so move SCLK to U44 and MISO to... no: keep U44 as drawn and use ISO7742F for ENABLE + MOSI only if SCLK can share U44's INC with GPIO moved to a jumper. Decide at order time.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - SPI Isolator	File Name: interface-spi-isolator.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[10] Interface - SPI Isolator/		Reviewer:	Size: **A3**	Sheet: **10** of **13**

[11] Interface - DAC Command

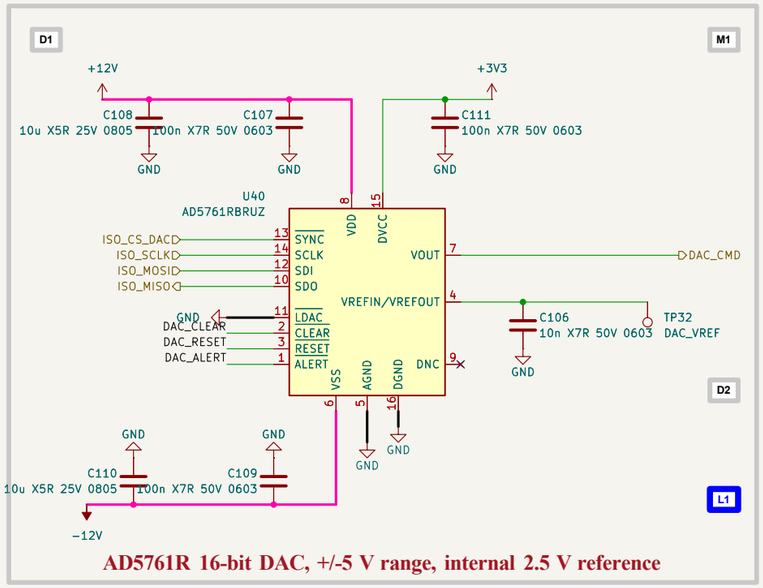
DESIGN NOTE D1:
AD5761R (16-bit, 8 software ranges, 2.5 V 2 ppm/C reference). Range +/-5 V -> DAC_CMD +/-5 V -> +/-5 mA at G = 1, so RG3 on [6] must be DNP when the DAC drives the channel (RG3 fitted = G = 5 is for the Red Pitaya +/-1 V outputs). Alternative kept in software: range +/-3 V (RA = 100) with RG3 fitted. 1 LSB = 153 uV = 0.15 uA at G = 1 [H]. Output amp drives 2 kOhm / 1 nF [DS]; RS1 1k + D3/D4 on [6] is the load.

DESIGN NOTE D2:
Supplies: VDD = +12V, VSS = -12V (bipolar range needs VDD 4.75-16.5 V, VSS -16.5-0 V) [DS]; DVCC = +3V3 so the logic matches the isolator side 2 and the FPGA in bypass. Decoupling drawn in the order it is laid out: rail -> 10 uF bulk -> 100 nF at the pin. VREFOUT: 10 nF to AGND per the pin table (not more: start-up time 3.5 ms [DS]). TP32 reads 2.500 V.

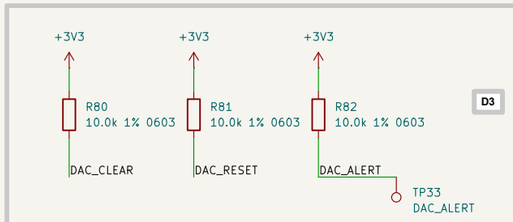
DESIGN NOTE D3:
Control pins: LDAC tied to DGND (DAC register updates when SYNC rises). CLEAR, RESET pulled to +3V3 (internal pull-ups exist; the 10k make the nets probe-able and let Rev 2 safety logic drive them). CLEAR (falling edge) sets the output to zero / mid / full scale per the control register: a candidate hard-wired clear from ISO_ENABLE low in Rev 2. ALERT open-drain, TP33.

LAYOUT NOTE L1:
C107 / C109 / C111 at pins 8, 6, 15 with the via at the pad, C108 / C110 next to them on the rail side; C106 at pin 4 with a short return to AGND. AGND and DGND on the same ground plane under the part. DAC_CMD short to RS1 on [6].

MISC NOTE M1:
SPI: 24-bit frames, SYNC low, data on SCLK falling edge, SCLK <= 50 MHz [DS]; write the control register first (range, CLR mode, overrange off), then input register 0x8000 = 0 V. Read-back: SDO drives ISO_MISO only while SYNC is low. Bring-up (R5 on the acceptance table): code sweep vs DMM on TP13 DAC_CMD. Sourcing checked 2026-11-07 [LCSC API / DigiKey CA web]: see U40 field notes.



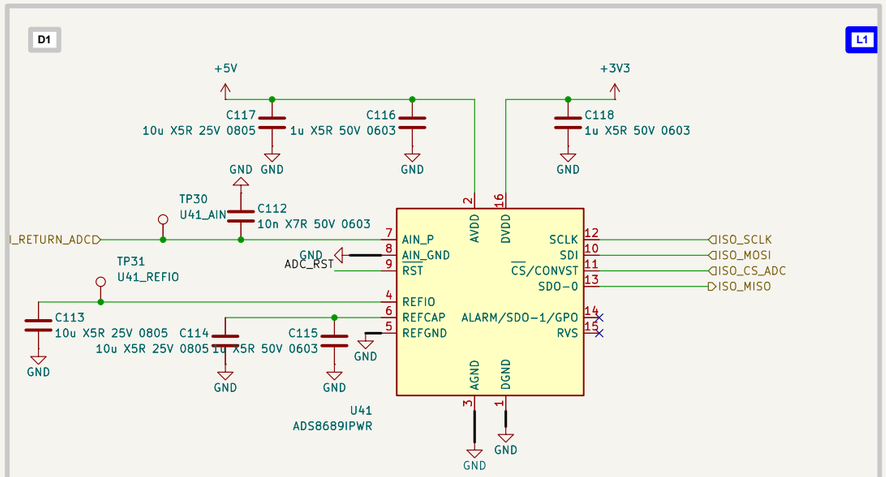
AD5761R 16-bit DAC, +/-5 V range, internal 2.5 V reference



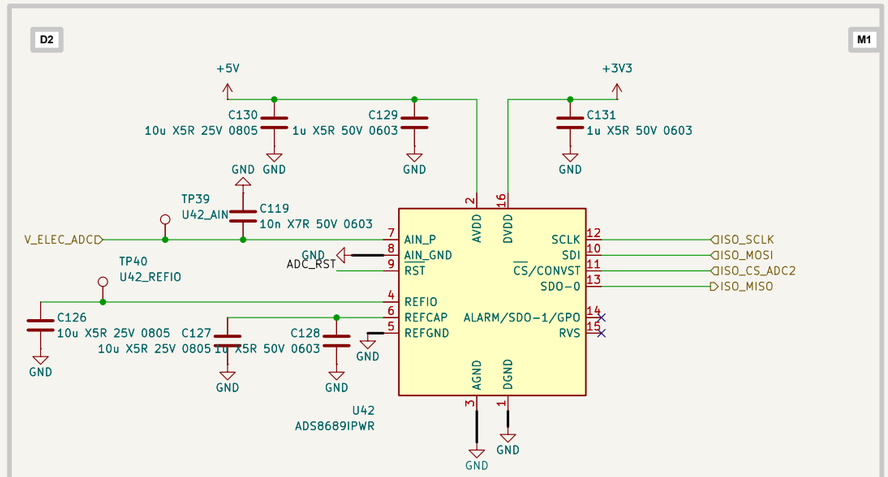
Control-pin pull-ups

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: **UBC ELEC 491 - JY063**		Variant:	Git Hash:
	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - DAC Command	File Name: interface-dac-command.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[11] Interface - DAC Command/		Reviewer:	Size: **A3**	Sheet: **11** of **13**

[12] Interface - ADC Read-back



ADC 1, U41: return current I_RETURN_ADC = VMID + 0.2 V/mA



ADC 2, U42: electrode voltage V_ELEC_ADC = VMID + V_elec / 101



Shared ADC reset pull-up

DESIGN NOTE D1:
Two ADS8689, one per read-back signal, on the shared isolated SPI with their own chip selects (ISO_CS_ADC on J3 pin 11, ISO_CS_ADC2 on J3 pin 15, both on the default-HIGH isolator U44). I and V are sampled in the same instant, there is no analog switch in the path and the firmware is the same driver twice (decision 2026-11-08, replaces the single ADC + CD74HC4052 mux). VMID_REF is a DC constant (LM4040 0.1 % x 1 % divider): measure it once on TP29 [9] and subtract in software; it is not digitised.

DESIGN NOTE D2:
Range 0-5.12 V (1.25 x VREF): VMID + 0.2 V/mA uses 13-52 % of full scale, 0.39 uA/LSB, noise-free ~13 bits [DS SNR 92 dB]; V_elec 7.9 mV/LSB. The input is a 1 MOhm resistive divider rated +/-20 V [DS], so a saturated OPA4192 on [9] needs no clamp. Rev 2 option: use the bipolar +/-2.56 V range and drop the VMID adder on [9].

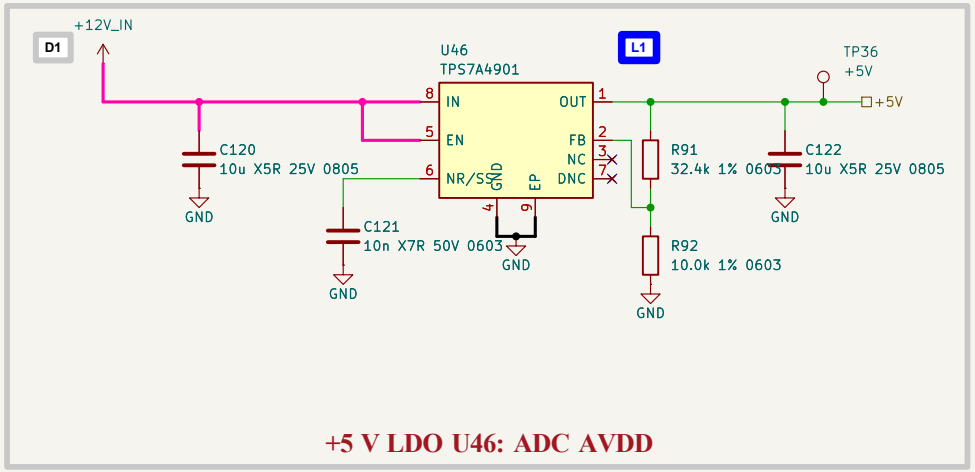
DESIGN NOTE D3:
Supplies drawn at the pins in layout order: +5V -> 10 uF -> 1 uF -> AVDD; +3V3 -> 1 uF -> DVDD [DS 8.3.1]. AVDD must be 4.75-5.25 V; DVDD could be 5 V as well, but 3.3 V keeps the SPI at Red Pitaya levels for the J5 bypass and matches the isolator side 2 and the DAC DVCC. Reference: internal 4.096 V, 10 uF on REFIO, 1 uF nearest REFCAP then 10 uF, returned to REFGND [DS]. RST shared, pulled up by R85; GPO / RVS not used (NC). SDO-0 is tri-state while CS is high, so both ADCs and the DAC share ISO_MISO [DS].

LAYOUT NOTE L1:
Per ADC: 1 uF then 10 uF at REFCAP (pin 6), 10 uF at REFIO (pin 4), returns to REFGND (pin 5) and the plane under the part; 1 uF at AVDD and DVDD, via at the pad; 10 nF across pins 7-8. Keep I_RETURN_ADC / V_ELEC_ADC from [9] away from the SPI traces; route the bus on the digital side of both parts. AGND, DGND, REFGND: one plane.

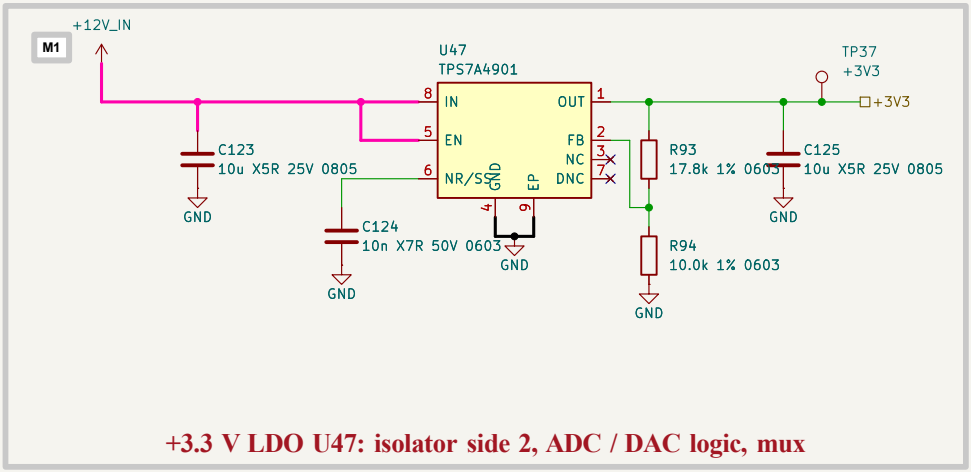
MISC NOTE M1:
Firmware: after reset write RANGE_SEL_REG = 0b1011 (0-1.25 x VREF) to both; conversions by CS low/high, 32-bit frame, data in the top 16 bits [DS]; never assert both chip selects at once. Read-back check R4: DMM on TP30 / TP39 vs code; TP31 / TP40 = 4.096 V. Sourcing checked 2026-11-07 [LCSC API / DigiKey CA web]: see U41 / U42 field notes.

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	Board Name: **STIM Rev 1 Channel**		Project Name: **JY063 STIM stimulator**	
Sheet Title: Interface - ADC Read-back	File Name: interface-adc-readback.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[12] Interface - ADC Read-back/		Reviewer:	Size: **A3**	Sheet: **12** of **13**

[13] Power - Converter Rails



+5 V LDO U46: ADC AVDD



+3.3 V LDO U47: isolator side 2, ADC / DAC logic, mux

DESIGN NOTE D1:
+5V and +3V3 are new rails for the converters and the isolator board side (added 2026-11-07). Both TPS7A4901 (as U30), fed from +12V_IN (the raw J2 input, same GND) rather than the A1212S +12V so U32's 83 mA budget is untouched [H: existing +12V load ~30 mA + 40 mA would be close]. FB: R91/R92 and R93/R94 by the D-01 method; 10 nF NR/SS, 10 uF X5R in and out as on sheet [4]. LCSC numbers for 32.4k / 17.8k 0603 to confirm (TBD in the fields).

MISC NOTE M1:
Bring-up order: these rails before anything digital (R5 in the acceptance table): TP36 = 5.04 +/- 0.1 V, TP37 = 3.30 +/- 0.07 V with J3 unplugged. +12V_IN now carries ~+40 mA extra: check the bench supply limit and D8.

LAYOUT NOTE L1:
Same rules as sheet [4] L1: PowerPAD (pin 9) to GND with vias, CIN / COUT / NR cap at the pins. Place U47 near the isolators (side 2) and U46 near U41.

Comments: PRELIMINARY: not for fabrication Edited in KiCad: do not regenerate Netlist: expected_netlist_rev1min.csv Not a safety sign-off; dummy loads	Company: UBC ELEC 491 - JY063		Variant:	Git Hash:
	Board Name: STIM Rev 1 Channel		Project Name: JY063 STIM stimulator	
Sheet Title: Power - Converter Rails	File Name: power-converter-rails.kicad_sch	Designer: WeymenKoo	Date: 28-Oct-2026	Revision: 2.1.0-dev
Sheet Path: /[13] Power - Converter Rails/		Reviewer:	Size: A4	Sheet: 13 of 13